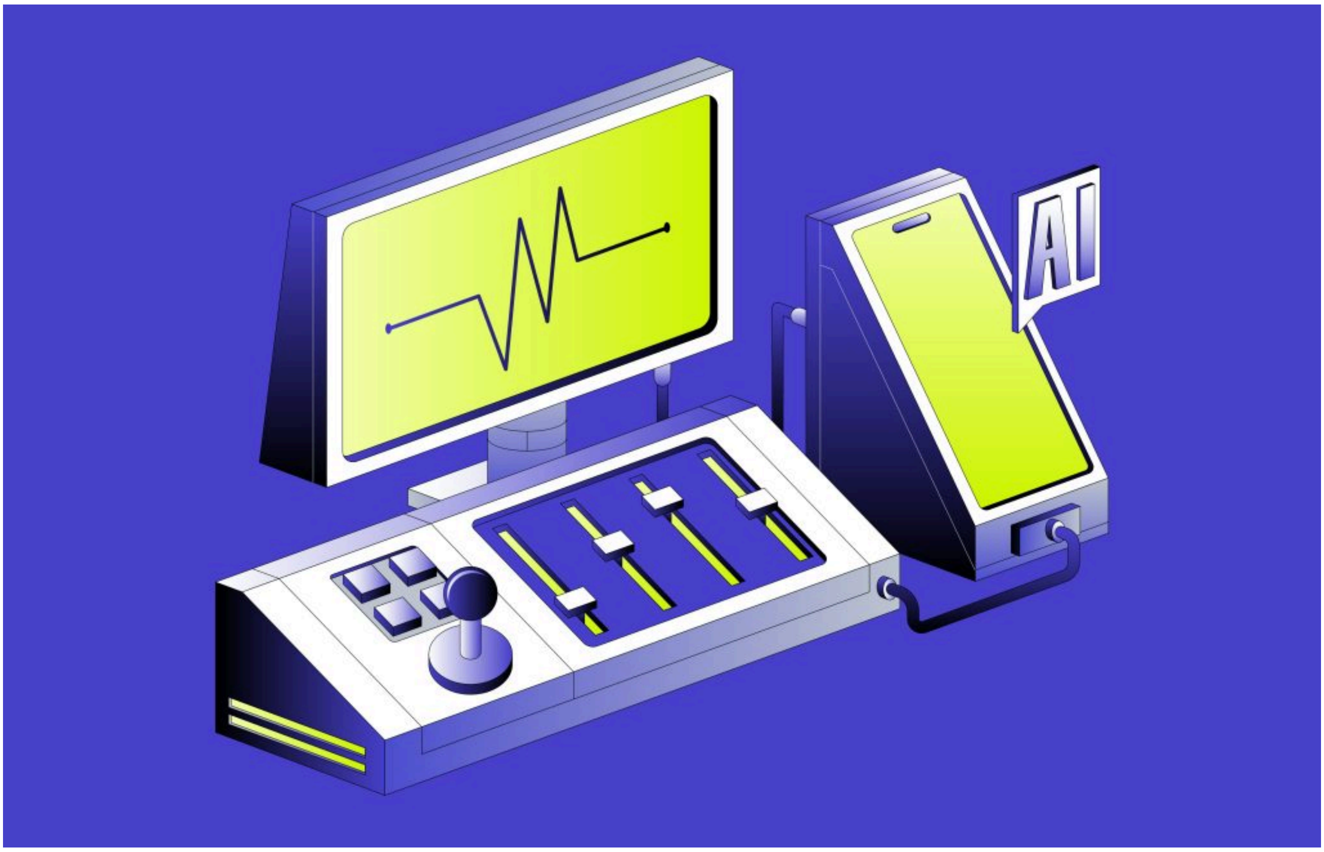
АГЕНТЫ ИИ / ИНФРАСТРУКТУРА ИИ / СЕТЕВЫЕ ТЕХНОЛОГИИ

По мнению Aria Networks, показатель Model Flop Utilization станет определяющим для эпохи инфраструктуры искусственного интеллекта.

Aria Networks запускает инициативу "Сеть, которая думает", направленную на оптимизацию производительности кластера искусственного интеллекта за счет использования гибкой модели, звука, телеметрии и агентов искусственного интеллекта.

7 апреля 2026 года, 9:00 Адриан Бриджуотер



флаг летнего времени

По мере того как ускоряется глобальная гонка за предоставление инфраструктурных услуг в области искусственного интеллекта, Aria Networks заявляет, что разработала «принципиально иной подход» к работе сетей и к ышению эффективности токенов в эпоху агентных технологий.

работает комбинация технологий и методологий, включая инструменты для оптимизации использования флоп-модели (MFU), недавно разработанную компанией Aria SONiC (сетевую операционную систему с открытым исходным кодом для центров обработки данных, оптимизированных для распределения), сквозную сверхтонкую телеметрию и интеллектуальные агенты, которые работают по всему сетевому стеку.

Что такое коэффициент использования модельных флопов?

Коэффициент использования модельных флопов, который компания Aria Networks называет «определяющим показателем» эпохи фабрик искусственного интеллекта, измеряет эффективность работы оборудования в центрах обработки данных по отношению к теоретически достижимой пиковой пропускной способности. Он может служить косвенным показателем того, окупаются ли вложения в ИИ-кластер.

MFU напрямую определяет эффективность токенов и стоимость одного токена. Поскольку токены становятся тем, что Ария называет «валютой интеллекта», эффективность инфраструктуры сети влияет на скорость синхронизации градиентов (математических сигналов, которые обновляют весовые коэффициенты модели), на эффективность передачи кэшей «ключ-значение» (чтобы модели не обрабатывали повторно предыдущие токены), а также на то, насколько гладко происходит планирование заданий на тысячах графических процессоров, тензорных процессоров, нейронных процессоров и т. д.

> «Если сеть не работает на полную мощность, то все остальные инвестиции в оптимизацию оказываются напрасными». — Мансур Карам, основатель и генеральный директор Aria Networks

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Сеть внутри кластера

Мансур Карам, основатель и генеральный директор Aria Networks, говорит, ... командам, отвечающим за работу сети, и инженерам-программистам ...обходимо понимать, почему их расходы на сеть (которые, по его оценкам, грань между успехом и провалом наиболее очевидна.

He makes this assertion because network teams can optimize the job scheduler, the storage layer, or the KV cache transfer algorithm, but each of those optimizations depends on an optimized network to achieve the expected result.

“Without the network performing at its best, the gains from every other investment are left on the table,” Karam tells *The New Stack*. “The Aria solution distinguishes between updates that affect the data plane, control plane, and management plane. Updates that affect the data plane are treated very differently from upgrades that only affect the management plane.”

Deliberately hybrid

He explains that his company has adopted a hybrid architecture, and deliberately so. The Aria agent layer spans multiple layers of the stack — from the switching ASIC layer (an Application Specific Integrated Circuit designed to route data packets in a specific way) up through the controller (where network traffic is configured and orchestrated), all the way to the cloud.

Throughout this hybrid architecture, different layers operate at different resolutions and with different methods and intelligence requirements. At the lowest levels, So the era of automated infrastructure continues to grow. This tier of the technology stack now provides everything from serverless functions to automated provisioning and self-healing instances with autonomous load balancing. Does that mean developers should consider their years spent getting an MSc in cloud communications & networking somewhat redundant? Perhaps Aria's hardened SONiC implementation could expose APIs for developers who want to build custom tooling or integrate their own networking constructs into existing infrastructure-as-code pipelines, right?

"While Aria Networks introduces a transformational operational model designed to maximize AI cluster utilization, it's built to be open and drop into existing environments and toolsets," says Karam.

He further explains that Aria's SONiC distribution preserves the standard interfaces that developers already rely on, so existing tooling continues to work without modification. The Aria platform also exposes a REST API, CLI, and MCP interfaces that developers can use to interact with the Aria Server to integrate "deep networking" (a branded term that Aria capitalizes to denote its work covering ultra-fine grain telemetry and deep network visibility) into existing infrastructure.

How fine-grained is fine-grained telemetry? In Aria's case, it's 10–10,000x finer resolution than traditional tools, collected across switches, transceivers, and hosts in a single unified view.

Agents partner alongside network engineers

At the operator-facing layer (Aria Console), the agents include leading LLM models. Operators primarily interact in natural language: they can ask questions about the network state, request explanations of alerts, and collaborate with the system on remediation decisions.

The LLM has access to the full telemetry and system state. It uses a specialized networking context, meaning its responses and actions are grounded in the accuracy, safety, and reliability standards required by network operators. The agents partner with the operators to enable continuous network optimization.

e champion an automated testing culture, whereby systems are continuously
d thoroughly tested 24/7 before any new updates are pushed out. Updates go

> **The bottom line on the use of networking agents working to devise strategies for resolving issues or optimizing performance is a simple pledge – this is not a black box; it is a partnership.**

As with every AI advancement, this technology is not about putting network engineers and operators out of a job; instead, it’s about enabling capabilities such as intent-based configuration. Network operators specify their needs, and the platform configures the fabric accordingly for routing, load balancing, congestion management, and failover. This is intended to reduce (Aria would say eliminate) the manual, error-prone workflows that slow down traditional deployments.

Karam promises transparency and control; his company’s bottom line on the use of networking agents working to devise strategies for resolving issues or optimizing performance is a simple pledge. This is not a black box; it is a partnership.

TNS

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